AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

MT141
Introduction to Probability and Statistics
Midterm Examination (MTA)
Fall Semester 2025/2026

Date: 15 / Nov. /2025

Number of Exam Pages: (3) (including this cover sheet)	**Time Allowed: 90minutes**

Instructions:

Please read the following instructions before starting:

- Total Number of Questions: 4
- This is a closed book exam.
- Be sure you write your name and ID on the External Answer booklet.
- This exam accounts to 30% of the total mark in the course.
- Budget your time according to the mark assigned to each question.
- Mobile phones and all other mobile communication equipment are NOT allowed.
- Calculators are allowed.

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Part 1: MULTIPLE CHOICE QUESTIONS

Solve all the questions. Each question is worth 1.5 marks, giving a total of 6 possible marks.

Q–1: If a pair of fair dice is rolled, what is the probability of getting a total of 9?

a) 1/12
b) 1/9
c) 1/6
d) 1/4

Q–2: A researcher recorded the number of cups of coffee consumed daily by a group of employees over a week. The recorded numbers (in hundreds) were: 8, 10, 12, 14, 16, 18, 20. What is the sample mean number of cups consumed per day?

a) 12
b) 13
c) 14
d) 15

Q–3: A warehouse ships two types of packages, Package A and Package B, using separate delivery services. The probability that Package A arrives on time is 0.9. The probability that Package B arrives on time is 0.8. Assuming the deliveries are independent, what is the probability that both packages arrive on time?

a) 0.17
b) 0.72
c) 0.10
d) 0.98

Q–4: In a standardized test, a student scored higher than 25% of all participants. Which percentile best describes this student's score?

a) 25th
b) 50th
c) 75th
d) 100th

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Part 2: ESSAY QUESTIONS

Each question is worth 8 marks. You should answer all the three questions, giving a total of 24 possible marks.

Q–1: [2+2+1+3 Marks]

A hospital administrator is evaluating patient flow efficiency during peak outpatient hours. To assess waiting time before consultation, they recorded the following wait times (in minutes) for a random sample of 15 patients:

8, 10, 7, 12, 9, 10, 11, 8, 13, 10, 9, 12, 7, 11, 10

a) **Calculate the sample mean for these data.**

b) **Calculate the sample median and mode.**

c) **Find the sample range.**

d) **Calculate the sample standard deviation.**

Q-2: [3+2+3 Marks]

A box contains **5 blue pens** and **3 black pens**. Two pens are randomly selected in succession **without replacement**.

(a) What is the probability that the **second pen selected is black**?

(b) What is the probability that **both pens selected are of the same color**?

(c) What is the probability that the **first pen selected is blue**, **given** that the second pen selected is black?

Q–3: [2+3+3 Marks]

A school administrator is analyzing the number of late arrivals in randomly selected classroom sessions. Let Y be a discrete random variable representing the number of students who arrived late in a given session. The probability distribution is given in the table below:

Y	0	1	2	3	4
P (Y=y)	0.1	0.3	L	0.2	0.1

a) **Find the value of L.** b) **Find $P(Y \leq 2)$.** c) **Find $P(Y > 1)$.**

End of questions

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